

Total no. of Pages: 01..	Roll no.
Degree: B.Tech.....	Semester: — Course work
MID-SEMESTER EXAMINATION (Summer), June 2026	
Course Title: Electronic Devices and Circuits	
Course Code: ECECC:202/EIECC:202/ICECC:202/VTECC:202	
Duration: 1:30 Hours	Max. Marks:20.....
Note: - Attempt all questions in the given order only. Missing data/ information (if any), maybe suitably assumed & mentioned in the answer.	

Q.No.	Question	Marks	CO
1a	Discuss intrinsic and extrinsic semiconductors with suitable energy band diagrams.	2	CO1
1b	What is the concentration of the holes in Si crystals having doner concentration of $1.4 \times 10^{24} / \text{m}^3$ when the intrinsic carrier concentration is $1.4 \times 10^{18} / \text{m}^3$? Find the ratio of the electron to hole concentration.		CO1
2a	Discuss drift current and diffusion current in semiconductors. Derive the expressions for drift current density and diffusion current density.	2	CO1
2b	The mobility of free electrons and holes in pure germanium are 3800 and 1800 $\text{cm}^2/\text{V-s}$ respectively. The corresponding value of pure silicon are 1300 and 500 $\text{cm}^2/\text{V-s}$ respectively. Determine the value of intrinsic conductivity for both. Assume $n_i = 2.5 \times 10^{13} \text{ cm}^{-3}$ for germanium and $n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$ for silicon at room temperature.	2	CO1
3a	Define Fermi level and derive the expression for carrier concentration in intrinsic and extrinsic semiconductors.	2	CO1
3b	Explain the formation of a p-n junction. Discuss the depletion region, barrier potential, and V-I characteristics of a p-n junction diode.	2	CO2
4a	Describe the construction, operation, and characteristics of Zener diode. Explain its application as a voltage regulator.	2	CO2
4b	A Zener regulator diode has: Input Voltage = 20 V, Zener Voltage = 10 V, Series Resistance = 500 Ω Load Resistance = 1 k Ω Determine: Load Current, Series Current, Zener Current and Zener Power Dissipation	2	CO2
5a	Explain the operation of emitter follower, BJT with suitable circuit diagrams.	2	CO3
5b	For a silicon transistor CE circuit: $V_{CC} = 12 \text{ V}$, $R_C = 2 \text{ k}\Omega$, $R_B = 470 \text{ k}\Omega$, $\beta = 100$, $V_{BE} = 0.7 \text{ V}$ Calculate: Base current I_B , Collector current I_C , Collector-emitter voltage V_{CE} and Determine the region of operation.	2	CO3